NILAM — Layered System Architecture

Nilam Client Application
React + Google Maps

Requests

Layer 1: Input & Evidence Ingestion

User Input Gateway
Site · Radius · Text · Voice · House Brief

Geospatial Data Sources
Earth Engine · OSM · Kerala Hazards

Location Services
Maps · Places · Street View

Evidence Preprocessor
38 Features · Seasonal Sentinel-2 Chips

Prepared evidence

Layer 2: Land Context Processing

Candidate Grid & Site Selection
Point · Polygon · Radius Search

38-Feature Engine
Terrain · Soil · Climate · Access

Safety & Open-Land Gates
Slope · Flood · Landslide · Buildings

Preference Router
Laya + Distil-Whisper

Features

Satellite chips

Layer 3: Multimodal Land Suitability Intelligence

TabPFN 3.5
Tabular Site Evidence
92% Fusion Weight

TerraMind Base
Dry + Monsoon Sentinel-2
8% Fusion Weight

Probability Fusion
Four Suitability Classes
Final Suitability Percentage

SHAP Explanation
Feature Drivers
Evidence Summary

Land Results
Score · Ranked Flat Land
Amenities · Street View

Selected land

Layer 4: ArchAI2 House Generation

Site Context
Location · District
Suitability Score

Qwen 3.5 9B
House Requirements
Local LM Studio

Spatial Specification
Rooms · Floors · Setbacks
Adjacency · Plot Geometry

Layout Generation
HouseGAN++ Proposals
Procedural Optimizer

Geometry Validation
Access · Area · Circulation
Collision · Setbacks

Design Outputs
2D Plan · 3D Model
Revisions · Exports

Layer 5: Kerala Cost Estimation & Delivery

Saved Design Geometry
Measured Floor Area

Kerala Cost Engine
District · Finish · Allowances · Reserve

Final Product Outputs
Land Report · House Design · Budget Range · Cost Breakdown · Exports

Results and interactive views

Model artifacts

Project state

Persistent Data, Model Registry & Project State

1,400-Point Statewide Cohort · Seasonal Satellite Chips · TabPFN / TerraMind Artifacts · Fusion Metrics

SQLite Projects & Revisions · Canonical House Geometry · Feature / Satellite Caches